MECH 191 — Metallurgy

Semester Metallography Portfolio — Instructor Answer Key

EXAMET-5-R-600-HD · 50–500× Reflected Light Microscopy

Fall 2026 · Leeward Community College · Instructor: Kai Santos

How to Use This Key

This document mirrors the structure of MECH191_Modules.docx — the Safety & PPE Quiz, then Modules 1–15 in order. It is written to guide grading, not to be handed to students.

For measurement-based questions (grain diameter estimates, Vickers indent diagonals, ASTM E112 grain-size numbers), there is no single fixed numeric answer — each student measures their own photograph. Grade the **method** (correct formula/procedure applied) and check that the reported number falls in the plausible range noted for that specimen; treat a value far outside that range as a flag to double-check the student's measurement or arithmetic, not automatically as wrong.

For short-answer and critical-thinking questions, award credit for responses that reach the same substantive conclusion with sound reasoning, even if the wording differs from the model answer below.

The portfolio's own Grading Summary (reproduced below) governs point allocation: 5 pts/module × 15 modules = 75 pts, plus a 25-pt holistic end-of-semester review = 100 pts total.

Grading Summary

Component	Points	Notes
Observation table	1 pt	Credit for each feature located and checked
Sketch / photograph	1 pt	Labeled with specimen, magnification, etchant
Short-answer questions (Step 3)	1 pt	Credit apportioned across all questions in the set — see model answers below
Critical-thinking questions (Step 4)	1 pt	Equation applied correctly and reasoning tied to the observed microstructure
Kit return & handling	1 pt	On time, no damage, handled by edges
Total per module	5 pts	× 15 modules = 75 pts
End-of-semester portfolio review	25 pts	Holistic; all 15 modules as a complete document
TOTAL	100 pts	

Safety & PPE Quiz — Answer Key (Complete Before Module 1)

Passing score: 100% (18 of 18 correct).

Section A — Etchant & Chemical Hazard Identification

Q1. (b) Nital 2%. Per the Etchant SDS page, Nital's diluent is methanol (86–91%), not the ethanol commonly assumed for a "nital" recipe.

Q2. (b) Keller's Reagent and Kroll's Reagent. Both SDS documents list hydrofluoric acid (≈1% and 1–2% respectively) as a component.

Q3. (b) Apply calcium gluconate gel. Both SDS documents specify calcium gluconate gel as the skin first-aid response for HF exposure — it binds free fluoride ions before they can penetrate tissue and pull calcium from the blood.

Q4. (b) Danger. All six etchant SDS documents (Nital, Marble's, Kalling's No. 2, Keller's, Kroll's, Ferric Chloride) carry the signal word "Danger."

Q5. (b) It does not reliably block hydrofluoric acid. HF passes through latex; the PPE guide specifies nitrile or neoprene gloves — never latex — for Keller's and Kroll's reagents.

Q6. (c) Kalling's No. 2 Reagent. Its SDS lists H226 (flammable liquid & vapor), H314 (severe skin burns/eye damage), and H331 (toxic if inhaled) together. (Nital also carries all three hazard classes but is not one of the answer choices — among the four options given, only Kalling's No. 2 has all three.)

Q7. (b) In a fume hood. The PPE guide states HF-containing reagents (and Nital, for its methanol vapor) must be mixed and etched only in a fume hood.

Q8. (b) Optic nerve damage / risk of blindness. The Etchant SDS page calls out methanol's H370 hazard (damage to organs, specifically the optic nerve) as distinct from the other diluents used in the lab (ethanol, water).

Section B — Match Each Etchant to the Specimen Material It's Used On

Each letter is used exactly once; the correct pairings (confirmed against every module's stated etchant/specimen combination) are:

#	Etchant	Correct Answer	Specimen Material
1	Nital 2%	**B**	Carbon steel (1018, 1045, 1075)
2	Marble's Reagent	**F**	Austenitic stainless steel (304, 316)
3	Kalling's No. 2	**C**	Precipitation-hardening stainless (17-4PH)
4	Keller's Reagent	**D**	Aluminum (6061, 6063, 5052, 2024)
5	Ferric Chloride	**E**	Brass and copper
6	Kroll's Reagent	**A**	Titanium (Ti-6Al-4V)

Section C — PPE Selection

Q9. (b) ANSI Z87.1 safety glasses. Required at all times in the shop per the PPE guide's General Shop PPE baseline — even when only observing.

Q10. (b) A full face shield. The PPE guide specifically calls out Kroll's reagent as requiring a full face shield in addition to (never instead of) chemical splash goggles.

Q11. (b) Wear no gloves at all. Gloves must be removed before operating any machine with an accessible rotating or moving part (lathe, mill, drill press) — the PPE guide flags this as the one place in the shop where removing PPE is the correct safety action.

Q12. (c) Nitrile or neoprene. Required for mixing/handling any of the six lab etchants; never latex.

Q13. (b) Calcium gluconate gel. Must be staged at the etching station any time Keller's or Kroll's reagent is in use.

Q14. (d) 12–14. Per the Welding Helmet Lens Shade Chart, SMAW (stick) above 160 A calls for shade 12–14 (60–160 A calls for 10–12).

Section D — Short Answer / Applied Scenarios

Q15. Two PPE errors: (1) **Latex gloves** — latex does not reliably block hydrofluoric acid, so HF can pass through to the skin, where it causes painless-at-first deep tissue damage and can pull calcium out of the bloodstream; nitrile or neoprene gloves are required instead. (2) **No face shield** — Kroll's reagent specifically requires a full face shield in addition to eye protection because it stops splash to the whole face, not just the eyes; goggles/glasses alone leave the rest of the face exposed to an HF splash. (A strong answer may also note the "basic safety glasses" are not the chemical splash goggles required for etchant work — full credit either way as long as two distinct, correctly explained errors are given.)

Q16. Gloves are removed at a lathe or drill press because rotating/moving machine parts (chuck, spindle, bit) can catch loose glove material and pull the hand in before the operator can react — this has caused documented finger and thumb amputations. In a chemical or thermal context a glove is a barrier that protects the hand; at rotating machinery a glove becomes the hazard itself, so the correct action is to remove it rather than add it.

Q17. Calcium gluconate gel must be on hand any time Keller's or Kroll's reagent (both HF-containing) is in use. It treats HF skin exposure: the gluconate ion binds free fluoride ions in the tissue before they can continue penetrating and leaching calcium out of the blood and underlying bone, which is what makes untreated HF exposure potentially fatal over a large enough contact area.

Q18. The blanket "safety glasses and gloves" rule is not sufficient because different etchants demand different tiers of the same PPE category. For example, HF-containing Keller's and Kroll's reagents require nitrile/neoprene gloves specifically (not just "gloves" — latex fails to block HF), chemical splash goggles rather than basic safety glasses, and Kroll's additionally requires a full face shield; work with any etchant must also happen in a fume hood. A student following only the general rule could reach for latex gloves and basic glasses and be seriously under-protected for an HF exposure.

Module Answer Keys 1–5

Grading note: Questions that ask students to measure something from their own photograph/sketch (grain diameter, indent diagonal, ASTM grain-size number) do not have one fixed numeric answer — grade the method and whether the reported number falls in a plausible range for that specimen (ranges noted below). Everything else below is a full model answer.

MODULE 1 · WEEK 1 — Atomic Structure & Bonding (Specimen A, 1018 steel)

Step 3

Q1. At 100×, 1018 (normalized) shows a roughly **equiaxed** grain boundary network — grains are approximately equal in all directions, consistent with a normalized (air-cooled from above the upper critical temperature) low-carbon steel that has not been significantly deformed afterward.

Q2. Metallic bonding is **non-directional**: the positive ion cores are held together by a delocalized "sea" of shared electrons rather than fixed, directional bonds (as in covalent/ionic solids). Because the bonding isn't tied to a specific bond angle or neighbor, atoms at a grain boundary — where the lattice is disordered and misoriented relative to each neighboring grain — can still be held together by that electron sea. The boundary is higher-energy than the bulk lattice but not a broken bond, so the material doesn't crack simply because a boundary exists.

Q3. Method: count the number of grains crossing the ~2.3 mm field width at 100× and divide 2.3 mm by that count. A typical count for normalized 1018 is roughly 12–20 grains across the field, giving an average grain diameter in the range of **≈0.12–0.19 mm**. Accept any answer that shows the correct division and a plausible count for the student's own photo.

Q4. At 500×, **MnS inclusion stringers** become visible — small gray, elongated particles aligned with the rolling direction that are too fine to resolve at 100×.

Step 4 — Critical Thinking

C1. Sample X (finer grain) is expected to have the **higher yield strength**. Hall-Petch ($\sigma y = \sigma 0 + ky/\sqrt{d}$) shows yield strength increases as d decreases. Physically, more grain boundary area per unit volume means more obstacles that block and pile up dislocations at boundaries; a higher applied stress is needed to push dislocations across more, closer-spaced boundaries and activate slip in the next grain.

C2. No — strengthening cannot increase without limit as $d \to 0$. In real metals, once grain size drops into the nanocrystalline range (roughly below ~10–15 nm), there isn't enough room inside a grain to pile up dislocations at all, and deformation shifts to grain-boundary sliding and grain-boundary diffusion instead of dislocation motion — this actually causes strength to level off or even decrease again (the "inverse Hall-Petch" effect). Processing limits on how fine a grain size can actually be produced are a practical limit as well.

MODULE 2 · WEEK 2 — Mechanical Properties (Specimens A 1018, B 1045)

Step 3

Q1. Specimen B (1045) shows **more pearlite** than A (1018). 1045 has roughly 0.45% C versus 1018's ~0.18% C; by the lever rule, higher carbon content increases the equilibrium pearlite fraction in a normalized hypoeutectoid steel.

Q2. Method: measure both diagonals of the Vickers indent on B at 100× (field width ≈2.3 mm), average them (in mm), then apply $HV = 1.854 \times 9.81 / d_{avg}^2$. Grading check: normalized 1045 typically reads in the **~170–210 HV** range — flag results far outside that band for a possible measurement or arithmetic error, not necessarily a wrong method.

Q3. Halving the grain diameter increases the $k_y/\sqrt{d}$ term by a factor of **$\sqrt{2}$ (≈41%)**. Since $\sigma_y = \sigma_0 + k_y/\sqrt{d}$, total yield strength increases by less than 41% (because σ_0 doesn't change), but the strengthening contribution from grain refinement alone grows by $\sqrt{2}$.

Q4. Specimen B (1045) is expected to have the **higher tensile strength**, and the microstructure supports it: more pearlite means more of the harder, stronger ferrite-cementite lamellar phase and less of the softer, more ductile ferrite.

Step 4 — Critical Thinking

C1. The **ferritic** specimen shows more strain at the same engineering stress. Ferrite has a lower yield strength than pearlite, so it yields and begins plastically deforming at a lower stress; by the time both specimens reach the stated stress level, the ferritic one has already accumulated more plastic strain, while the stronger, less ductile pearlitic specimen is still closer to its own yield point. Microstructure — not stress alone — sets each material's stress-strain path.

C2. Past the yield point, dislocations move and multiply on slip planes, producing **permanent (plastic) atomic-plane displacement** rather than the small, reversible elastic stretching of atomic bonds that governs Hooke's law. Once slip occurs, removing the load does not return the atoms to their original relative positions, so stress and strain are no longer linked by a single elastic constant — the relationship becomes governed by dislocation density, motion, and interaction (strain hardening) instead of bond stretching.

MODULE 3 · WEEK 4 — Carbon & Alloy Steels (Specimens A 1018, B 1045, C 1075)

Step 3

Q1. A correct schematic places all three points on the room-temperature (ferrite + cementite) side of the diagram, below the eutectoid line, with: 1018 (~0.18% C) far to the left of the eutectoid composition (0.76% C, mostly ferrite with a small amount of pearlite); 1045 (~0.45% C) roughly midway between 0 and the eutectoid composition (a visible ferrite + pearlite mix, close to even); 1075 (~0.75% C) essentially at the eutectoid composition (nearly fully pearlitic).

Q2. Ranking by pearlite fraction: **A (1018) < B (1045) < C (1075)**. This matches the lever-rule prediction — higher carbon content directly increases the calculated pearlite fraction, and it should match what's visually darker/more pearlite-dominant under the scope.

Q3. Pearlite lamellar spacing is controlled primarily by **cooling rate**, not composition — all three specimens were normalized under similar conditions, so spacing should be broadly similar across A, B, and C even though their pearlite *fractions* differ substantially.

Q4. No — hardness should **not** scale linearly (4×) with carbon content. Hardness tracks pearlite fraction, and the lever rule shows that relationship is nonlinear and bounded (fraction pearlite can't exceed 100%, reached at the eutectoid composition, 0.76% C). Ferrite also still contributes a baseline hardness in every specimen. So while 1075 is harder than 1018, the increase is much smaller than proportional to the 4× carbon increase.

Step 4 — Critical Thinking

C1. Increasing carbon from 1018 to 1045 **increases** the pearlite fraction — the lever-rule numerator ($C0 - 0.022$) grows as C0 increases toward the eutectoid composition. This should match the microscope observation of visibly more dark pearlite in Specimen B than A.

C2. Above 0.76% C (hypereutectoid), the simple hypoeutectoid lever-rule equation no longer applies. Instead of "more than 100% pearlite," the actual microstructure develops a **proeutectoid cementite (Fe_3C) network** along prior austenite grain boundaries in addition to pearlite — a different two-phase tie line (austenite + cementite) governs that composition range, which this equation doesn't capture.

MODULE 4 · WEEK 5 — Stainless Steel Families (Specimens D 304, E 316, F 17-4PH)

Step 3

Q1. Annealing twins are visible in D and E because austenitic (FCC) stainless steels have **low stacking fault energy**, which makes it energetically easy to form twin boundaries during recrystallization/annealing. Specimen F is martensitic (body-centered tetragonal), formed by a diffusionless shear transformation rather than annealing/recrystallization, so it does not develop annealing twins by that mechanism.

Q2. Molybdenum improves resistance to **pitting and crevice corrosion in chloride environments**, largely through an electrochemical effect on the passive chromium-oxide film's stability rather than a change in bulk crystal structure or grain morphology — so the ~2–3% Mo addition in 316 doesn't show up as a visible microstructural difference under the scope even though it meaningfully changes corrosion performance.

Q3. Specimen F shows a **martensitic lath/plate matrix with fine, dispersed copper-rich precipitates** from aging — no annealing twins, a completely different appearance from D and E. This illustrates that "stainless" is defined by composition (≥~10.5% Cr, forming a passive film) and corrosion behavior, not by any one crystal structure or microstructure — stainless steels can be austenitic, ferritic, martensitic, duplex, or precipitation-hardening.

Q4. 17-4PH would be expected to be **less prone to work hardening** during machining than 304. Its strength comes from precipitation hardening in a martensitic matrix, not from the FCC structure that gives austenitic stainless steels (like 304) their notoriously high work-hardening rate during machining.

Step 4 — Critical Thinking

C1. A 2–3% Mo addition is worth roughly **3.3 × 2–3 ≈ 6.6–9.9 PREN points**. Mo is weighted more heavily than Cr in the formula because, per weight percent, it is disproportionately effective at stabilizing the passive film specifically against chloride-induced pit initiation and repassivation — a targeted, high-leverage effect beyond Cr's general passive-film formation.

C2. PREN is a compositional formula calibrated primarily for austenitic and duplex stainless steels; applying it to 17-4PH is less reliable because a martensitic, precipitation-hardened microstructure can have different local corrosion behavior (e.g., around precipitates or prior-austenite boundaries) that bulk composition alone doesn't capture. PREN is still a rough compositional indicator across families, but it's a weaker comparison once the base microstructure itself changes.

MODULE 5 · WEEK 6 — Nonferrous Metals & Phase Diagrams (Specimens G 6061-T6 Al, K 70/30 Brass)

Step 3

Q1. Specimen K (brass) shows **more prominent twins** than Specimen D (304 SS, Module 4), even though both are FCC. Brass has an even **lower stacking fault energy** than austenitic stainless steel, making twin formation still more favorable — wider, more numerous twin bands per grain.

Q2. Aging precipitates fine **Mg_2Si particles** (via GP zones and intermediate transition phases) that are coherent or semi-coherent with the aluminum matrix. These fine particles impede dislocation motion (particles must be sheared or bowed around, per the Orowan mechanism), which raises the alloy's hardness and strength.

Q3. The fine dark particles visible at 500× are **Mg_2Si precipitates**. They strengthen the alloy by acting as obstacles that dislocations must bow around or cut through, requiring higher applied stress for continued plastic deformation.

Q4. Solution treatment dissolves Mg and Si fully into a single-phase supersaturated solid solution, so that the subsequent aging step can precipitate Mg_2Si in a fine, evenly distributed, coherent form. If solution treatment is skipped, Mg and Si remain in coarse, non-uniform as-cast particles; aging without first solutionizing produces little strengthening because there's no supersaturated solid solution to precipitate finely from.

Step 4 — Critical Thinking

C1. The lever rule requires a **two-phase equilibrium region on a phase diagram with a tie line connecting the compositions of the two phases present** at a given temperature — any two-phase field (not just steel's ferrite + cementite) meets this requirement, so the Fe-Fe_3C case used in Module 3 is one specific application of the same general rule.

C2. The lever rule only tells you the **relative amounts (weight fractions) and compositions of the phases present at equilibrium** — it says nothing about precipitate size, spacing, coherency, or distribution, which is what actually controls strengthening in both precipitation hardening (aluminum) and solid-solution strengthening (brass). In other words, it tells you *how much* of each phase exists, not *how the phases are arranged* — and arrangement is what mechanical properties depend on.

Module Answer Keys 6–10

MODULE 6 · WEEK 7 — Mechanical Testing (Specimens A 1018, B 1045)

Step 3

Q1. Method: measure both Vickers indent diagonals on B, average them, apply $HV = 1.854 \times 9.81 / d_{avg}^2$ (d in mm). Grading check: normalized 1045 typically reads **~170–210 HV**.

Q2. Method: draw a test line across the 100× field (≈2.3 mm), count grain-boundary intersections, convert to grains/mm and then to ASTM Grain Size Number G using the E112 procedure (repeat over 3 lines and average, per Module 8). Grading check: normalized 1018 typically falls around **ASTM G 6–8**; accept any value the student's own intercept count supports, and check it's roughly consistent with their Module 1 grain-diameter estimate.

Q3. Using the student's own measured HV for B and UTS ≈ 3.3 × HV: this should land in the neighborhood of the given 750 MPa (e.g., an HV of ~200 predicts ≈660 MPa). Full credit for showing the calculation and a reasonable qualitative judgment (typically "roughly consistent, within the range expected for an empirical correlation" — exact agreement isn't expected).

Q4. Toughness would be expected to **increase** if grain size is refined. Unlike most strengthening mechanisms, grain refinement improves both strength *and* toughness together — more grain boundaries deflect and interrupt crack propagation, absorbing more energy before fracture (this is why grain refinement is often called the only strengthening mechanism that doesn't trade ductility/toughness for strength).

Step 4 — Critical Thinking

C1. Because d is squared in the denominator, measuring it 10% too large makes the calculated HV **too low**, by roughly a factor of $1/1.1^2 \approx 0.83$ — about **17% low**.

C2. The UTS ≈ 3.45×HV multiplier reflects steel's specific relationship between indentation resistance and tensile deformation behavior (elastic modulus, strain-hardening exponent, and how steel work-hardens under a tensile load). Aluminum and brass have different elastic moduli, crystal structures, and strain-hardening behavior, so the same empirical proportionality constant does not transfer — applying it would give a systematically wrong UTS estimate for those alloys.

MODULE 7 · WEEK 8 — NDT Methods (Specimens A 1018, L 360 leaded brass, P Ti-6Al-4V)

Step 3

Q1. Of the three, **UT (ultrasonic testing)** is the only one physically capable of detecting an internal, subsurface reflector like the lead particles in Specimen L — brass is non-ferromagnetic, so MT does not apply at all, and PT only detects flaws that break the surface. But even UT cannot distinguish *why* a reflector exists: it only reports that an acoustic-impedance mismatch is present, along with its size and location — not composition or intent. Confirming that a given reflector is an intentional lead addition rather than an unwanted inclusion requires correlating the UT signature with known material specifications or, ultimately, destructive metallography — which is exactly the limitation this question is illustrating.

Q2. Expect a **weak (or absent)** MT indication. MnS stringers are non-metallic inclusions that are often subsurface rather than surface-breaking, and MT sensitivity drops off quickly with depth — it's strongest for indications at or very near the surface. The stringer's elongation parallel to the rolling direction (and typically parallel to the surface) also makes it harder to interrupt magnetic flux lines effectively compared to a defect oriented perpendicular to the field, further weakening any indication.

Q3. Titanium aerospace components are flight-critical, expensive, and fatigue-sensitive; sectioning one for metallography permanently destroys it and removes it from service. NDT lets the same part be inspected repeatedly, non-destructively, throughout its service life, which is exactly what's needed for routine airworthiness monitoring. Destructive metallography is reserved for situations where understanding the actual microstructure is worth losing the part — root-cause failure investigation, or one-time qualification of a new process/material before it's approved for production.

Q4. Two justified situations: (1) **failure investigation** — after an in-service crack or fracture, to determine the root cause and confirm the failure mechanism, which requires seeing the actual microstructure; (2) **process/material qualification** — sacrificing sample parts from a new fabrication method, heat lot, or supplier to verify the resulting microstructure and properties meet spec before releasing the process for production.

Step 4 — Critical Thinking

C1. Using $\lambda = v/f$, detecting a smaller defect requires a **higher-frequency** transducer (smaller $\lambda \rightarrow$ smaller dmin). The trade-off is reduced penetration depth — higher-frequency sound attenuates faster in the material, so the inspector loses the ability to reach deep sections in exchange for finer near-surface resolution.

C2. Two specimens could produce a very similar UT signature (same reflector size/location, similar amplitude) while being completely different under the microscope — for example, a small gas pore, a nonmetallic inclusion, and a different-phase constituent particle can all create a comparable acoustic-impedance mismatch despite being metallurgically unrelated. Likewise, grain size or texture differences that are obvious optically may not create enough acoustic contrast to register differently in a UT signal at all.

MODULE 8 · WEEK 9 — Quality Standards & Defect Recognition (Specimen A, 1018)

Step 3

Q1. Method: three test lines at 100×, count intersections per line, average, and convert to ASTM Grain Size Number G per E112. Grading check: expect a result broadly consistent with the student's Module 6 measurement (typically **G 6–8** for normalized 1018).

Q2. Expected table for 1018 bar stock: **Type A (sulfide/MnS stringers) — Thin** (light stringers, consistent with the inclusions seen since Module 1); **Types B (alumina), C (silicate), D (globular oxide) — none observed / not applicable**, since clean normalized carbon-steel bar stock like this specimen typically shows only Type A sulfide inclusions. Accept any answer that correctly applies the comparator-chart method even if the specific ratings differ slightly.

Q3. Compare the student's own G number (Q1) to G7 (~32 µm grain diameter). If their measured grain size is coarser than G7 (a lower G number / larger measured diameter), the specimen does **not** meet a "G7 minimum" callout; if finer (higher G number), it does. Normalized 1018 often runs slightly coarser than G7, so many students may correctly conclude the specimen does not meet the spec — grade on whether the comparison and conclusion are logically consistent with their own G number, not on a single fixed answer.

Q4. Any four of: specimen identification (material grade, heat/lot number); etchant and magnification used; test method/standard referenced (e.g., ASTM E112) and the numerical result with units; date of examination; inspector name/initials or signature; equipment used/calibration reference.

Step 4 — Critical Thinking

C1. In $N = 2^{(n-1)}$, a **higher** ASTM grain size number n corresponds to **smaller** grains (more grains per square inch). Reconciling with Hall-Petch: smaller grains raise yield strength, so a higher ASTM grain size number corresponds to a **stronger** material.

C2. Reproducibility matters more in a quality-standards context because the result is a pass/fail compliance decision that different inspectors, labs, and time periods must agree on consistently — an "eyeball estimate" introduces subjective variation that could pass a nonconforming part or fail a conforming one. In a research setting, the goal is usually to characterize a general trend, where some measurement variability is acceptable and can be handled statistically rather than needing a single defensible, auditable number.

MODULE 9 · WEEK 10 — Wrought Aluminum Alloy Systems (Specimens G 6061, J 2024)

Step 3

Q1. Specimen J (2024, Al-Cu) is expected to show **coarser, more numerous** constituent particles than G (6061, Al-Mg-Si). This matches expectation — Al-Cu system intermetallics (e.g., Al_2Cu, Al_2CuMg) typically form coarser, more numerous constituent particles than the finer Mg_2Si-based constituents typical of 6xxx alloys.

Q2. Copper-rich constituent particles in 2024 act as local galvanic cathodes relative to the surrounding aluminum matrix, promoting localized/pitting corrosion at those particle sites; copper in solid solution also disrupts the uniformity of the passive oxide film. 6061's Mg-Si-based intermetallics are far less galvanically active, giving it better inherent corrosion resistance — which is exactly why 2024 is often sold Alclad (clad in a thin layer of pure aluminum) for corrosion protection.

Q3. No — swapping aging schedules would **not** be expected to give similar strengthening in both. The Al-Cu precipitation sequence (GP zones → θ″ → θ′ → θ) and the Al-Mg-Si sequence (GP zones → β″ → β′ → β) are chemically distinct, with different optimal aging times/temperatures tuned to each system's own precipitation kinetics. Applying the wrong schedule risks under-aging or over-aging, producing sub-optimal strength in either alloy.

Q4. Beyond corrosion resistance: 2024 (aerospace skin alloy) generally reaches a **higher peak strength/strength-to-weight ratio** suited to fatigue-critical aircraft structure, while 6061 offers better weldability and more forgiving general-purpose fabrication — a reasonable engineer's answer citing either strength or weldability/fabricability is acceptable if the reasoning ties back to the observed constituent-particle differences.

Q5. Method: measure and calculate HV for both. Grading check: typical values are roughly **6061-T6 ≈ 95–105 HV** and **2024-T4 ≈ 120–135 HV**; expect J (2024) to read harder, consistent with its more numerous/coarser strengthening constituents observed in Q1.

Step 4 — Critical Thinking

C1. Both d (grain diameter) and L (precipitate spacing) represent the **characteristic spacing between obstacles to dislocation motion** — grain boundaries in one case, precipitate particles in the other. Both appear as an inverse (or inverse-square-root) term in their respective strengthening equations because smaller spacing means a dislocation encounters obstacles more often as it moves, requiring more stress to keep deforming the material.

C2. Between a well-aged and a slightly under-aged alloy, expect to see (1) **precipitate size** — larger, more fully developed precipitates in the well-aged condition versus very fine, sub-critical GP zones/transition precipitates when under-aged — and (2) **precipitate spacing/density**, which shifts as precipitates grow and coarsen toward the aging peak. Both are visible under the microscope at 500×.

MODULE 10 · WEEK 11 — Aluminum Alloys for Forming Applications (Specimens H 6063, I 5052)

Step 3

Q1. At 500×, Specimen H (6063) shows **fine Mg_2Si precipitates**, reflecting its heat-treatable Al-Mg-Si chemistry; Specimen I (5052) shows **no discrete precipitates** — just a uniform matrix with Mg retained in solid solution, since 5052 is not heat-treatable.

Q2. 6063 carries lower overall alloy content than 6061, giving a simpler, more homogeneous microstructure with fewer/smaller constituent particles. Fewer hard particles mean lower resistance to metal flow through the extrusion die, less die wear, and a smoother, more polishable surface finish — exactly why 6063 is the standard architectural extrusion alloy.

Q3. A cold-stamped 5052 sample would be expected to show **elongated/deformed grains** in the direction of working and evidence of increased dislocation density (e.g., more strongly etching deformation bands near the surface) rather than any change in precipitates — since 5052 strengthens by work hardening, not aging.

Q4. **5052** is the better choice for a marine-application bracket: as a non-heat-treatable Al-Mg alloy it has excellent corrosion resistance in chloride/marine environments (no copper-based galvanic activity) and good sheet formability for stamping. 6063 is optimized for extrusion, not sheet forming, and is a less natural fit for a stamped part.

Q5. Method: measure and calculate HV for both. Grading check: 6063 and 5052 typically land fairly close to each other (roughly **HV 60–75**), and both read noticeably softer than the higher-alloy-content 6061/2024 values logged in Module 9 — despite reaching their strength through different mechanisms (precipitation vs. work/solid-solution hardening), neither is a high-strength alloy relative to 6061/2024.

Step 4 — Critical Thinking

C1. A higher strain-hardening exponent n means the material keeps gaining strength locally as it deforms, which spreads strain more evenly across the part rather than letting it localize. In a stamping operation this delays localized necking/thinning and tearing at the most-stretched region of the part, which is critical for a deep-drawn or stamped shape like 5052 sheet.

C2. This shows that "formability" is not a single property. Extrudability (6063) is about how uniformly material flows through a die under hot compressive/shear conditions — governed mainly by alloy content and flow-stress uniformity — while stamping performance (5052, via strain-hardening exponent n) is about room-temperature tensile ductility and resistance to localized thinning. An alloy optimized for one forming process is not automatically well-suited to the other, because each process stresses a different aspect of the material's deformation behavior.

Module Answer Keys 11–15

MODULE 11 · WEEK 12 — Machining Processes (Specimens A 1018, B 1045, G 6061)

Step 3

Q1. Yes — expect a thin **near-surface deformation layer** at 500× on Specimen A's prepared edge: grains flattened/smeared and elongated parallel to the surface from grinding/polishing-induced plastic deformation, distinct from the equiaxed grains seen in the interior.

Q2. Increased cutting force/heat from tool wear can locally heat a high-carbon steel workpiece above its critical temperature at the very surface, followed by rapid self-quenching from the surrounding cold mass — producing a thin, hard, featureless **untempered ("white") martensitic layer** at 500× that etches very differently (lighter, harder to etch) than the surrounding pearlite/ferrite structure. Lower heat inputs could instead produce a locally tempered/overtempered zone.

Q3. Any clearly identified surface feature that concentrates stress — e.g., a preparation/machining scratch, or an MnS inclusion intersecting the surface — is a valid **fatigue crack initiation site**: sharp surface features act as local stress risers where a fatigue crack can nucleate under cyclic loading.

Q4. **Aluminum (Specimen G)** typically shows more preparation-induced deformation. It has a lower yield strength/hardness than the steels, so it deforms (smears) more readily under the mechanical action of grinding and polishing.

Q5. Using representative logged values (G ≈ 95–105 HV, A ≈ 130–150 HV, B ≈ 170–210 HV), the ranking softest → hardest is **G < A < B**. This predicts B (1045) would need the **lowest cutting speed** and show the **most tool wear** for a given tool life, while G (6061 aluminum) — the softest — would tolerate the highest cutting speeds with the least tool wear.

Step 4 — Critical Thinking

C1. From $V \cdot T^n = C$, increasing cutting speed V **decreases** tool life T. A harder specimen (higher HV) generates more cutting force and heat at a given speed, accelerating tool wear — so it needs a **lower** cutting speed to achieve the same tool life as a softer specimen.

C2. The Taylor equation says nothing about **microstructural features like abrasive hard-phase content or built-in chip-breaking constituents** (e.g., inclusion morphology, free-machining particles such as lead in brass, or pearlite lamellar structure). Two specimens with identical bulk HV can still machine very differently if one contains hard abrasive particles that accelerate tool wear, or soft discrete particles that promote favorable chip breaking — bulk hardness alone doesn't capture this.

MODULE 12 · WEEK 13 — Copper Alloys: Strengthening Mechanisms (Specimens N 110 Cu, O 182 Chromium Copper)

Step 3

Q1. Specimen N looks visually **"simpler"** — just grain boundaries and annealing twins, no precipitates — because pure copper has no solute atoms available to segregate or precipitate. Specimen O, with chromium in solution, develops a more complex microstructure once aged.

Q2. Common thread across 17-4PH (steel, Cu-rich precipitates), 2024 (aluminum, Al_2Cu/Al_2CuMg precipitates), and chromium copper (Cr-rich precipitates): all three reach their strength through the same **solutionize → quench → age** sequence, producing a fine dispersion of a second-phase precipitate that blocks dislocation motion (the Orowan mechanism) — the base metal and precipitate chemistry differ, but the underlying strengthening mechanism is identical.

Q3. Any alloying addition — even a small amount — disrupts the periodic regularity of the host lattice, and conduction electrons scatter off that disorder (impurity/solute scattering, per Matthiessen's rule). This scattering happens regardless of how small the addition is, so even a modest chromium addition measurably raises resistivity compared to pure copper.

Q4. A PM (sintered) part would be expected to show **porosity/voids** between former powder particles — a microstructural feature (isolated or interconnected pores) not present in any of the fully dense wrought specimens examined this semester.

Q5. Method: measure and calculate HV for N and O. Grading check: annealed pure copper typically reads **~40–50 HV**; aged chromium copper typically reads **~110–130 HV** — roughly a 2.5–3× jump. This relative gain is closer in scale to the aluminum family's precipitation-hardening gain (Module 9) than to steel's much larger 17-4PH gain (Module 14) — accept reasoned comparisons based on the student's own measured numbers.

Step 4 — Critical Thinking

C1. By Matthiessen's rule (ρtotal = ρthermal + ρimpurity), at the same test temperature ρthermal is equal for both specimens, but O has an added **ρimpurity** term from chromium solute/precipitate scattering that N lacks — so O's total resistivity is higher.

C2. The strengthening benefit isn't free because the same microstructural feature — fine chromium-rich precipitates/solute atoms — is responsible for **both** effects simultaneously: it blocks dislocation motion (raising strength via the Orowan mechanism) **and** scatters conduction electrons (raising resistivity via Matthiessen's rule). The two effects can't be decoupled because they share the same physical cause.

MODULE 13 · WEEK 14 — Brass Alloys: Composition and Microstructure (Specimens K 260, L 360, M 464)

Step 3

Q1. Annealing twins are clearly visible throughout Specimen K's single-phase alpha matrix. In L, twins may still be present within the alpha matrix but are visually harder to pick out amid the scattered dark lead particles; in M, twins are confined to the alpha regions only (not present in the beta phase), so the two-phase structure reduces overall twin visibility/density compared to K.

Q2. Lead is essentially insoluble in the Cu-Zn matrix, so it stays as discrete, soft particles rather than dissolving into (and strengthening) the matrix. During machining, these soft particles shear easily and act as internal chip-breakers/lubricant pockets, causing chips to break into short segments instead of forming long, stringy chips — exactly the insolubility that makes lead "useless" as an alloying strengthener is what makes it effective for machinability.

Q3. M containing two phases means its zinc content **exceeds the solubility limit of Zn in the copper-rich alpha solid solution** (roughly 35–38 wt% Zn on the Cu-Zn diagram) at room temperature, pushing its composition to the right of the alpha solvus and into the alpha+beta two-phase field — while K's lower zinc content (~30%, cartridge brass) stays within the single-phase alpha field.

Q4. This is a genuine trade-off, not a clean win for either alloy: **360 (L)** machines the best of the three thanks to its lead addition but has no specific dezincification protection; **464 naval brass (M)**'s beta phase is actually the more zinc-rich, more chemically reactive phase and is itself preferentially attacked by dezincification — in real naval brass, a small tin addition (not modeled by this simplified three-specimen, tin-free comparison) is what actually suppresses that beta-phase attack. Based only on what's observable here, a defensible answer picks M for its recognized marine-service microstructure while explicitly flagging that its beta phase is a corrosion liability, not a guarantee — full credit for either K/L/M as long as the beta-phase dezincification trade-off is correctly identified and reasoned through.

Q5. Method: measure and calculate HV for K, L, and M. Grading check: single-phase K and L typically read close together (**~55–80 HV**), while duplex M typically reads somewhat **higher (~75–90 HV)** due to the harder beta phase — consistent with phase count/composition predicting relative hardness.

Step 4 — Critical Thinking

C1. For a second (beta) phase to appear at all, M's zinc content must exceed the **maximum solubility of Zn in the copper-rich alpha solid solution** at that temperature — i.e., its composition must fall to the right of the alpha solvus line into the two-phase alpha+beta field on the Cu-Zn diagram.

C2. No — lead does **not** appear anywhere in the binary Cu-Zn lever-rule calculation; it's a third element entirely outside that two-component system (and is essentially insoluble, so it doesn't participate in the Cu-Zn phase equilibrium at all). This shows the fundamental limit of a two-component phase diagram: it can only account for the two elements it plots. Any third or further alloying addition's effect on the real alloy — like lead's machinability boost — has to be understood separately, since simple binary lever-rule math cannot capture ternary (or higher) alloy behavior.

MODULE 14 · WEEK 15 — Precipitation Hardening: A Cross-Family Comparison (Specimens F 17-4PH, J 2024, O 182 CrCu)

Step 3

Q1. Expected table:

Specimen	Base metal	Precipitate phase	Appearance at 500×
F	Steel (martensitic)	Copper-rich (ε-Cu) clusters	Fine, hard to resolve
J	Aluminum (FCC)	Al_2Cu / Al_2CuMg	Coarser constituents clearly visible, plus finer strengthening precipitates
O	Copper (FCC)	Chromium-rich particles	Fine, hard to resolve

Q2. **Solutionize** — heat into the single-phase field so the alloying element(s) fully dissolve into solid solution. **Quench** — cool rapidly enough to trap that solute in a supersaturated, non-equilibrium solid solution at room temperature (preventing slow, uncontrolled formation of coarse equilibrium particles). **Age** — reheat in a controlled way (or hold at room temperature for natural aging) to let the supersaturated solution precipitate a fine, evenly dispersed second phase, which is what actually blocks dislocations and raises strength. (Example: in 2024, solutionizing dissolves Cu into the aluminum matrix, quenching traps it there, and aging precipitates fine Al_2Cu/Al_2CuMg particles.)

Q3. A fine, closely spaced precipitate field forces a dislocation to bow more sharply (smaller radius of curvature) to bypass each particle, which requires higher applied stress — this is the Orowan mechanism. A coarse, widely spaced precipitate field gives dislocations more room to bow around with comparatively little resistance, so for the same volume fraction of second phase, finer/closer-spaced precipitates are always more effective obstacles.

Q4. Not necessarily — natural aging at room temperature relies on diffusion that is far slower than at elevated aging temperatures. Some alloys (2024 does naturally age meaningfully) can approach a natural-aged condition given enough time, but the precipitation sequence favored at room temperature isn't always the same one that produces peak strength — natural aging can stall at an early transition-phase stage (e.g., GP zones) without ever forming the more effective intermediate precipitate that artificial aging reaches in hours, so the final strength reached is often lower and can take years rather than hours.

Q5. Method: measure HV for F; pull logged HV for J and O. Using the given baselines (304 ≈150 HV, 6061 ≈30 HV, pure Cu ≈40 HV) as each family's annealed reference point: F (17-4PH, aged, typically ~350–400 HV) gains roughly **+200–250 HV** over its steel-family baseline; J (2024, typically ~120–135 HV) gains roughly **+90–105 HV** over its aluminum-family baseline; O (chromium copper, typically ~110–130 HV) gains roughly **+70–90 HV** over its copper-family baseline. **Steel (F) shows the largest relative strength gain from precipitation hardening**, with aluminum and copper roughly comparable to each other and both well behind steel.

Step 4 — Critical Thinking

C1. No — a larger G alone does not guarantee the largest $\Delta\sigma$, because Orowan strengthening ($\Delta\sigma \approx Gb/L$) also depends on **L, the interparticle spacing** (and b, which varies less across metals). A very fine, closely spaced precipitate field in a lower-G metal could still out-strengthen a coarser, more widely spaced precipitate field in a higher-G metal. You'd need each alloy's actual measured precipitate spacing to know for certain.

C2. Based on the Q5 hardness data, steel (F) does show the largest gain, consistent with what its higher G alone would predict — but that agreement doesn't prove G is the only factor at work. If F's Cu-rich precipitates also happen to be finer/more closely spaced than J's or O's precipitates, then L is reinforcing G's advantage rather than working against it, and the HV ranking alone can't separate the two effects — a full answer should note that the data are consistent with G dominating, but that precipitate spacing differences could equally be inflating (or offsetting) that ranking.

MODULE 15 · WEEK 16 — Course Capstone & Synthesis

Step 3

Q1. Example (any correct specimen/family per crystal structure is acceptable): **BCC** — Specimen A, 1018 steel (ferrite, body-centered cubic at room temperature). **FCC** — Specimen D, 304 austenitic stainless steel (or 316/E, or the brass/copper specimens). **HCP** — Specimen P, Ti-6Al-4V titanium (Module 7) — predominantly HCP alpha phase at room temperature.

Q2. FCC metals like 304, 316, and brass have **low stacking fault energy**, which makes it energetically easy for partial dislocations to form a low-energy twin boundary during annealing/recrystallization. The BCC carbon steels examined this semester don't share that low-stacking-fault-energy annealing-twin mechanism — BCC metals can form mechanical (deformation) twins under specific conditions (shock loading, very low temperature, high strain rate), but not the simple annealing twins seen in the FCC specimens.

Q3. Example using Specimen O (182 chromium copper): **precipitation/age hardening** is the dominant mechanism — fine chromium-rich precipitates block dislocation motion (Orowan mechanism) after a solutionize-quench-age treatment. **Grain-boundary (Hall-Petch) strengthening** also contributes to some degree, since O still has ordinary grain boundaries like every specimen examined. Solid-solution strengthening is a minor contributor at best, since chromium's solubility in copper is low. (Any specimen is acceptable if the mechanism(s) claimed match its actual composition/processing.)

Q4. Example chain, citing at least three specimens: **Unit 1 (atomic structure/bonding)** — FCC vs. BCC crystal structure explains why annealing twins appear in Specimen D (304, FCC) but not Specimen A (1018, BCC). This crystal structure and bonding, together with alloy design, determines **Unit 2 (alloy family/composition)** — carbon content controls the ferrite/pearlite balance across A, B, and C (1018/1045/1075). Those resulting phases are what **Unit 3 (mechanical testing/inspection)** measures directly — the Vickers hardness logged for Specimen B correlates with its measured pearlite fraction. That same composition-property relationship drives **Unit 4 (manufacturing/alloy selection)** — Specimen H (6063) is deliberately kept lower in alloy content than 6061 specifically for extrudability. Finally, composition also governs **Unit 5 (service behavior/corrosion)** — Specimens D and E (304 vs. 316) show how a small molybdenum addition raises PREN and chloride resistance without changing the base microstructure. (Specimens A/B/C, D/E, and H are cited above — any three consistent, correctly reasoned specimens earn full credit.)

Q5. Compiling representative logged HV values (softest → hardest): **N (~45) < K (~60) < I (~65) < H (~68) < L (~75) < M (~82) < G (~100) < O (~120) < J (~127) < A (~140) < B (~190) < F (~375)**. Example adjacent pair from different families: **K (260 brass, alpha solid solution) and I (5052 aluminum, Al-Mg solid solution)** land close together even though they're different base metals, because both derive their strength from the same basic mechanism — solid-solution strengthening only, with no precipitation and no second phase — and solid-solution strengthening tends to produce a broadly comparable, moderate hardness increment across many alloy systems. (Accept any correctly reasoned adjacent pair supported by the student's own compiled log.)

Step 4 — Critical Thinking

C1. Because PREN is a purely **compositional/electrochemical** formula describing passive-film stability against chloride attack, it doesn't touch the Hall-Petch or precipitation strengthening pathways at all. 316 gets essentially the same grain structure and dislocation-blocking behavior as 304 (same annealing twins, same austenitic microstructure, Module 4) — the added molybdenum only changes the passive film's chemistry. This is why an engineer can re-engineer a metal's corrosion resistance (adding Mo) without touching its strengthening mechanism or mechanical properties at all — the two are largely independent design levers.

C2. **Hall-Petch** connects most directly to what's visible under the EXAMET-5: grain diameter is something students literally measured with the intercept method (Modules 1, 6, 8) and plugged straight into $\sigma y = \sigma 0 + ky/\sqrt{d}$. **PREN** is the hardest to connect to a visual observation — Module 4 showed that molybdenum's corrosion benefit produces no visible microstructural signature at all (304 and 316 look essentially identical under the scope), so PREN can only be inferred from known composition, never observed directly. (Orowan is a reasonable alternate answer for "hardest to connect," since precipitate spacing is often near or below practical resolution at 500×.)